

SHOVAN SHAKYA

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🐙 [momoisgoodforhealth.github.io/](https://github.com/momoisgoodforhealth)

Skills

Languages: C, C++, Python, Verilog

Software Development Tools: Zephyr RTOS, OpenCV, CUDA, Vivado, ROS2, CMake, Git

CAD Tools: Altium, KiCAD

Hardware & Debug Tools: Oscilloscope, Logic Analyzer, JTAG, SWD

Experience

REEKON Tools

Boston, Massachusetts

Embedded Software Engineer

June 2025 - Current

- Developed Zephyr RTOS firmware in C for the T1R Range hybrid measuring system on a Nordic nRF54 MCU, building the Kickstarter prototype firmware and maturing it toward production readiness for a product with \$300K+ in pre-orders.
- Implemented peripheral drivers (UART for LDM (Laser Distance Measure), stepper motors, I2C for battery charger, GPIO expander, magnetic sensor, IMU, SPI for TFT LCD).
- Improved SPI display driver refresh rate from 8 Hz to 30 Hz using semaphore based synchronization for tear-free LCD updates; implemented adaptive brightness control via ALS feedback while maintaining readability.
- Optimized T1R Range power draw across active, idle, and sleep modes via low-power firmware states and peripheral sequencing, display voltage rail tuning, reducing power draw from 112 mW to 61 mW.
- Added Bluetooth HID support to stream measurements as keyboard input.
- Implemented dual-axis leveling from accelerometer data with 30 Hz update rate and LPF for stable visualization.
- Developed an Embedded Linux + STM32 based LDM scanning prototype integrating camera sensors, stepper motors and quadrature encoder feedback for planar measurements.
- Developed multithreaded C++ image processing and motor control pipelines to detect laser dots, steer stepper motor to target using field oriented control (FOC) with PID based position control.
- Achieved sub-second inverse kinematics solver (previously 30 seconds) by engineering a custom C++ math engine to replace Python bottlenecks, leveraging Pybind11 and Eigen for zero copy memory integration, SIMD hardware acceleration.
- Developed Linux kernel device driver for LDM module.

Jaycon Systems

Palm Bay, Florida

Firmware Engineering Intern

September 2024 - May 2025

- Enhanced STM32 production firmware in C by mitigating SPI EMI issues for FCC compliance, adding ADC based temperature logging, and implementing SPI double buffering to eliminate DMA buffer corruption.
- Exposed sensor data (thermistor, VIN, GPIO) to a desktop application via C++ with SWIG generated C# bindings for diagnostics and monitoring.
- Contributed to a Zynq-7000 SoC image processing and state estimation system for a NASA SBIR project, implementing connected component labeling (CCL) in C++ on ARM Cortex-A9 (15 Hz at 720p) and developing FPGA-accelerated pipelines using C++ HLS and SystemVerilog with dual-port BRAM integration.

NBCUniversal

Orlando, Florida

Sensor Fusion Intern

May 2024 - August 2024

- Developed STM32 firmware in C for IMU interfacing over I2C, communicating with other STM32 modules via CANBUS, and transfer IMU data to a computer using UDP via Ethernet SPI.
- Designed a 4-layer STM32-based PCB (2cm x 2cm) with BNO088 IMU, Ethernet SPI module and TJA1051 CAN IC, supporting up to 30V input voltage.
- Developed a Python tool for synchronized pose and camera data capture on a NVIDIA Jetson based ROS2 platform.

Monterey Bay Aquarium Research Institute

Moss Landing, California

Computer Vision Intern

June 2023 - August 2023

- Developed OpenCV based realtime disparity and distance estimation tool for fisheye stereo cameras with accuracy range of 5 meters (+- 300 mm) for desktop and VR.
- Used multiprocessing to perform disparity and tracking in parallel.
- Integrated FathomNet YOLOv5 deep sea organism tracking model with distance estimation.
- Used Sockets to send realtime frames, and distance information between Python OpenCV application and Unity VR.

Education

University of South Florida

Tampa, Florida

Bachelor of Science in Electrical Engineering

May 2025